

Minerva Priyadarsini

Ph.D. Candidate in Electrical Engineering (Thesis Submitted)

School of Electrical Sciences, Indian Institute of Technology Goa

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ORCID | Google Scholar | GitHub

Expected Ph.D. defense: November 2026 | Available for postdoctoral positions thereafter

Research Interests: Next-Generation Wireless Communication · RSMA · SCMA/NOMA · MIMO · Multiuser Detection & Receiver Design · Wireless Signal Processing

Research Profile

Ph.D. candidate in wireless communications specializing in advanced multiple access, finite-alphabet communications, and physical-layer signal processing. Research spans RSMA/SCMA integration, MIMO-OFDM systems, codebook optimization, receiver design, and performance analysis under channel and hardware impairments. Published in *IEEE Transactions on Communications*, with recent work presented at IEEE PIMRC 2026.

Research Expertise

Multiple Access	Rate-Splitting Multiple Access (RSMA), Sparse Code Multiple Access (SCMA), hybrid RS-SCMA, code-domain NOMA, interference management, and high-overloading communication systems.
Transmission & MIMO	OFDM and wideband multi-carrier transmission, MIMO and MISO systems, IRS-assisted communication, subcarrier mapping, frequency diversity, and high-mobility communication.
Codebook Design & Optimization	SCMA codebook design and construction, differential evolution, and joint optimization of sum-rate and error-rate performance.
Detection & Receivers	Message-passing algorithm (MPA) based detection, successive interference cancellation (SIC), multi-user detection, and interference-aware receiver design.
Performance Analysis	Finite-alphabet error-rate and spectral-efficiency analysis under fading, interference, carrier-frequency offset, Doppler shift, time-selective channels, and hardware impairments; Monte Carlo link-level evaluation.

Education

2026	Doctor of Philosophy (Ph.D.) School of Electrical Sciences, Indian Institute of Technology Goa <i>Thesis: Sparse Code Multiple Access: From Codebook Design to Rate-Splitting Integration</i> <i>Thesis submitted; expected defense: November 2026</i>	Thesis Submitted
2018	M.Tech. in Communication and Signal Processing International Institute of Information Technology Bhubaneswar	CGPA: 9.15
2015	B.Tech. in Electronics and Communication Engineering Silicon Institute of Technology, Bhubaneswar	CGPA: 8.23

Research Experience

2019–2026	Doctoral Candidate , Indian Institute of Technology Goa - Proposed and analyzed the Rate-Splitting Sparse Code Multiple Access (RS-SCMA) framework, published in <i>IEEE Transactions on Communications</i>. - Extended RS-SCMA to OFDM and investigated subcarrier mapping, frequency diversity, Doppler, and carrier-frequency offset. - Designed and optimized SCMA codebooks using differential evolution, including joint sum-rate and error-performance optimization for MIMO-SCMA. - Analyzed SCMA-based systems under channel and hardware impairments using finite-alphabet analysis and Monte Carlo simulation.
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Publications

Journal Publication

- [1] **M. Priyadarsini**, Z. Liu, K. Deka, S. K. Sahoo, and S. Sharma, “Hybrid Rate-Splitting and Sparse Code Multiple Access (RS-SCMA): Design and Performance,” *IEEE Transactions on Communications*, vol. 74, pp. 7839–7855, 2026, doi: [10.1109/TCOMM.2026.3686697](https://doi.org/10.1109/TCOMM.2026.3686697).

Conference Papers

- [2] **M. Priyadarsini**, K. Deka, Z. Liu, and S. Sharma, “Rate-Splitting SCMA (RS-SCMA) over OFDM: Subcarrier Mapping and Frequency Diversity,” *IEEE International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC 2026)*, Singapore, September 2026.
- [3] **M. Priyadarsini**, K. Deka, Z. Liu, and S. Sharma, “MIMO-SCMA Codebook Design: Joint Optimization of Sum-Rate and Error Performance,” *IEEE International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC 2026)*, Singapore, September 2026.
- [4] **M. Priyadarsini**, K. Deka, Z. Liu, and S. Sharma, “Impact of Hardware Impairments on IRS-Assisted MISO-SCMA Communication Systems,” *IEEE International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC 2026)*, Singapore, September 2026.
- [5] K. Deka, **M. Priyadarsini**, S. Sharma, and B. Beferull-Lozano, “Design of SCMA Codebooks using Differential Evolution,” in *Proc. 2020 IEEE International Conference on Communications Workshops (ICC Workshops)*, Dublin, Ireland, 2020, pp. 1–7, doi: [10.1109/ICCWorkshops49005.2020.9145202](https://doi.org/10.1109/ICCWorkshops49005.2020.9145202).
- [6] **M. Priyadarsini** and C. Srinivasarao, “Beamforming in MIMO Radar using Coprime Array,” in *Proc. IEEE 2nd International Conference on Electronics, Materials Engineering and Nano-Technology (IEMENTECH 2018)*, India, May 4–5, 2018.
- [7] **M. Priyadarsini** and C. Srinivasarao, “Beamforming using Phased Coprime MIMO Radar,” in *Proc. IEEE International Conference on Inventive Communication and Computational Technologies (ICICCT 2018)*, India, April 20–21, 2018.

Teaching Assistant / Laboratory Instructor, Indian Institute of Technology Goa (2019–2024). Supported Digital Signal Processing, Control Systems, Probability and Random Variables, Digital Electronics, and Introduction to Basic Electrical and Electronics Engineering through laboratories, tutorials, quizzes, viva voce, and evaluation.

Awards & Grants

- **IEEE PIMRC 2026 Travel Grant Recipient**, IEEE PIMRC 2026, Singapore.
- **IEEE ICC 2020 Student Travel Grant Recipient**, IEEE International Conference on Communications (ICC 2020).
- **Best Master’s Thesis Award**, IIIT Bhubaneswar, for the M.Tech. thesis titled “*Beamforming and Performance Improvement in Co-located MIMO Radar.*”

Academic & Professional Service

- **Session Chair**, IEEE International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC 2026), Singapore, September 2026.
- **Technical Program Committee Member**, 7th Workshop on Emerging Topics in 6G Communications, IEEE GLOBECOM Workshops 2026.
- **Reviewer**, IEEE Transactions on Vehicular Technology

Technical Skills

- **Programming and Scientific Computing:** MATLAB and Python.
- **Communication-System Simulation:** Link-level simulation of RS-SCMA, SCMA, RSMA, OFDM, MIMO, and MISO communication systems; Monte Carlo performance evaluation.
- **Research Tools:** LaTeX, numerical optimization, performance analysis, and scientific visualization.